ECSPLAIN: Explainability-Constrained Classifier for Pairing the Detection and the Localization of Moving Areas From SAR Interferograms

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***Abstract*—Detecting slope instabilities on synthetic aperture radar (SAR) interferograms using deep learning approaches presents several challenges. This detection task suffers from the lack of transparency of deep networks, the complexity of the input data (i.e., complex values, sensitivity to distortions, and presence of counterfactuals), and the complexity of the target phenomena (i.e., the variable velocities and the complex underground processes). In this article, we propose a new framework called explainability-constrained classifier for pairing the detection and the localization of moving areas on interferograms (ECSPLAIN), to generate decision, localization, and segmentation maps from a single but explainable classifier network. It consists of training a classifier to detect whether an instability is located in the patch or not, and to explain its decision with a class activation map (CAM) that matches the actual location of the instability. Therefore, by using a single classifier network, the framework can pair the detection and the localization of moving areas. Four CAMs are investigated for the training of the ECSPLAIN framework. Experiments on the ISSLIDE dataset show that our proposal achieves better explainability than standard a posteriori CAMs with more than 0.20 points of improvement in terms of Dice and IoU scores. It also allows competitive performance with segmentation-only networks, with only 0.04 points of difference in terms of Dice and intersection over union (IoU) scores. Thus, the proposed method is competitive with the most efficient methods while being lighter, faster, and delivering a decision based on a human-like reasoning process. Finally, the ECSPLAIN framework is applied to enrich the ISSLIDE dataset, discovering more than 470 manually validated slope instabilities over the Alps.**

***Index Terms*—Deep learning, explainability, segmentation, slow-moving area, synthetic aperture radar interferometry (InSAR).**

I. Introduction

DEEP learning approaches have been widely exploited on remote sensing images for natural disaster monitoring [1], [2], detection [3], [4], and damage assessment [5], [6]. On the one hand, deep networks are leveraged for their fast inference and complex modeling capabilities, which allow a rapid response to disasters. On the other hand, the spatial coverage, the high acquisition frequency, and the plurality of satellite sensors allow the scaling of disaster monitoring across wide regions. This has already been implemented to detect sudden landslide collapses in [4], [7], and [8]. Nevertheless, the detection and localization of slow-moving instabilities are more challenging.

1) Slope instabilities can move very slowly (i.e., from centimeters to a few meters per year [9]), which limits the use of remote sensing images with decametric resolution such as those from Sentinel-1 and Sentinel-2. To overcome this issue, synthetic aperture radar (SAR) interferometry (InSAR) is exploited to detect centimeter-scale motions from the phase of SAR images [10], [11], [12], [13].
2) Slope instabilities are mostly located on steep slopes, which generate strong geometrical distortions in SAR images and interferograms [14].
3) The interpretation of InSAR data is difficult for non-experts because the images are complex-valued, show multiple counterfactual patterns, and are sensitive to external factors (e.g., atmosphere and topography).

In addition to the challenges posed by data and phenomena, neural networks are often considered as black boxes by non-experts in the field. This limits their understanding of the methods and lowers their confidence in the detections made by deep learning models in such challenging configurations. To better meet end users' expectations, recent researches propose explaining the behavior of these algorithms [15], [16] by using class activation maps (CAMs) [17].

CAMs are widely exploited in computer vision [18], [19], [20] to explain the decisions made by deep image classifiers. They generate a heat map that highlights the pixels that contribute the most to the decision based on the features extracted by the network. Such maps are often applied at inference time and are intended to guess the reasons for a given decision. Nevertheless, applying different CAMs to the same input-output pair can lead to incoherent explanations [21]. This issue is mainly due to the layer where the CAM is computed and the strategy used to combine feature maps (e.g., global or local weighting as detailed in Section IV-C). Therefore, intending to guess the reasons for a decision does not ensure that the explanation is correct since the network is

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neither forced to be explainable nor to comply with human explanation standards.

In this article, we propose a novel training strategy to force a classifier to make decisions based on a trustworthy and human-like reasoning. To this end, an explainability-constrained classifier for pairing the detection and the localization of moving areas on interferograms (ECSPLAIN) is introduced. This new framework proposes to leverage explainability strategies during the training of a ResNet50 classifier. The latter is trained on two tasks: 1) detecting the presence of slope instabilities in the input interferogram and 2) segmenting the moving region. The segmentation is computed by generating the CAM of the network at training time. Then, the CAM is constrained to match the true annotation map of the moving area. Four CAMs are explored to tackle this dual detection-localization task, namely Grad-CAM [18], Layer-CAM [22], NormGrad [21], and Score-CAM [23].

To the best of our knowledge, the proposed ECSPLAIN framework is the first method that constrains the explainability of a network to improve its accuracy in detecting slope instabilities on interferograms. First, it is easily understandable by end users, and it also achieves competitive performance with standard segmentation networks, while being lighter and faster than them. It is worth mentioning that the objective of this article is not to improve segmentation performance. Rather, we aim to encourage the development of explainability-based training strategies, which exploit end-users' expertise to guide the decision-making process of the network toward a human-like reasoning. The relevance of the proposed strategy is demonstrated through three experiments conducted on the ISSLIDE dataset[1] [13]. First, the ECSPLAIN framework is compared to segmentation networks to highlight its competitive performance with heavier and less understandable networks. Second, it is compared to a classifier trained without the explainability constraint to demonstrate the impact of the guidance on classification performance and on the relevance of extracted features. Finally, we demonstrate the applicability of the ECSPLAIN framework by generating localization maps and segmentation polygons of new slope instabilities in the French Alps. This results in the discovery of more than 470 instabilities, which were manually verified within a 49 500 km^2 area.

Therefore, the proposed contribution is threefold.

1) A novel ECSPLAIN framework is proposed to generate a least-cost segmentation map of moving areas on interferograms using a classifier network. This strategy shows competitive results with standard segmentation networks.
2) By constraining the explainability of the classifier, the ECSPLAIN framework forces it to make decisions based on human-like reasoning. This improves the relevance and the reliability of the detections quantitatively and qualitatively.
3) The ECSPLAIN framework is used to detect new moving areas on the interferograms available in the ISSLIDE dataset [13]. This results in the detection of more than 470 new moving areas, which were manually verified. Their segmentation and localization maps are included as additional annotation sources in ISSLIDE.

The remaining of this article is structured as follows. A general context and vocabulary clarifications are given in Section II. Related works on moving area identification and on CAM-based localization methods can be found in Section III. The ECSPLAIN approach and the selected CAMs are described in Section IV. The ISSLIDE dataset and the experimental setup are detailed in Section V before the presentation of results in Section VI. The ECSPLAIN framework is then applied to enrich the ISSLIDE dataset in Section VIII. Conclusions and perspectives end this article in Section IX.

II. General Context

A. Problem Statements

Due to the variety of definitions in the literature of geoscience and deep learning, this section clarifies the context of the current study.

1) Slope Instability/Moving Areas: In this article, both terms are used interchangeably and refer to slow-moving landslides [9] and rock glaciers [24]. The peculiarity of these phenomena is their very low and variable rate of movements (from cm/year to m/year within a single year [9]) without collapsing. Therefore, their detection and monitoring cannot be conducted with standard offset tracking techniques using remote sensing images such as Sentinel-1 or Sentinel-2 constellations due to their decametric resolution once projected on the ground. This issue is addressed in this article by using SAR interferometry as detailed in Section II-B.

2) Classification/Detection/Segmentation/Localization: In this work, the classification task is defined as assigning a label to a given input (e.g., "landslide," "rock glacier," or "motion free"). The detection task is considered more restrictive because it is a binary classification where the output of the model is a single bin. The absence (resp. presence) of a moving area should then result in a 0 (resp. 1) output. In the detection setup, both "landslides" and "rock glaciers" belong to the same class 1. The segmentation task consists of generating a map in which each pixel is assigned a label or binary detection value. Finally, the localization task consists of a fuzzy segmentation where the pixel values lie between 0 and 1. The greater the value, the more susceptible the pixel is in the class of interest.

3) Explainability/Interpretability: Geosciences are highly demanding in terms of lightness, fast computation, and understandability of the methods. Nevertheless, deep learning models are often considered as heavy "black boxes," where a huge quantity of parameters are used without an understandable reasoning process. To balance this debate and improve the confidence in neural networks, recent researches propose explanation or interpretation strategies [25]. Explaining a neural network consists of highlighting the features from the input that support the decision of the final classification/detection of the neural network (e.g., by generating heat maps to identify regions of interest in an input image). Conversely, the interpretability focuses on understanding the internal processing

[1]https://ieee-dataport.org/documents/isslide-insar-dataset-slow-sliding-area-detection-machine-learning

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mechanism of the neural network [26]. In this article, the focus is set on explainability and particularly on class activation maps (see Section III-B).

B. Interferometric SAR

This section offers a short reminder of SAR interferometry (InSAR) to facilitate the understanding of the choices made in the rest of this article.

Due to the development of on-demand services [27], InSAR data is highly favored over other modalities for detecting subsurface deformations (e.g., in buildings [28], [29], [30], volcanoes [1], [31], [32], or slope instabilities [10], [11], [13]). This process is based on the link between the phase φ of the complex SAR image $Ae^{i\varphi}$ and the travel time of the electromagnetic wave between the satellite and the ground. Therefore, if a motion Δd occurs in between two SAR acquisitions in the direction of the line of sight (LOS), it can be revealed by solving the following equation:

$$2 \times \Delta d = \frac{\lambda \times \Delta\varphi}{2\pi} \tag{1}$$

with λ the wavelength of the radar and $\Delta\varphi$ the phase difference between SAR images. Based on (1), if a 2π phase difference is observed in Sentinel-1 interferograms ($\lambda = 5.6$ cm), then a ground displacement of $\lambda/2 = 2.8$ cm occurred in the LOS direction.

The standard method for computing the phase difference map $\Delta\phi$ of two SAR images (z_1 and z_2) is to generate their complex interferogram as given in the following equation:

$$\rho e^{i\Delta\varphi} = \frac{\langle z_1 z_2^* \rangle}{\sqrt{\langle |z_1|^2 \rangle \langle |z_2|^2 \rangle}} = \frac{\sum_{n\in N} A_1^n A_2^n e^{i(\varphi_1^n - \varphi_2^n)}}{\sqrt{\sum_{n\in N} \left(A_1^n\right)^2 \times \sum_{n\in N} \left(A_2^n\right)^2}} \tag{2}$$

where ρ is named the coherence map, and A_i and φ_i refer to the amplitude and phase of z_i, respectively. N is the size of the neighborhood used in the multilooking process to estimate the mathematical expectation $\langle . \rangle$.

It is worth mentioning that the coherence map is of actual interest for the interpretation of interferograms, since it characterizes the reliability of the phase difference map.

III. Related Works

A. InSAR for Slow Motion Detection

Deep learning is widely exploited to process InSAR data for ground motion detection. A standard CNN architecture is trained in [33] to detect volcanic deformations based on interferograms computed by the LiCSAR processing tool [34]. Anantrasirichai et al. [31] also train a CNN to detect these deformations with an improved robustness to stratified and turbulent atmospheric patterns. Volcanologist expertise has been used to fine-tune the model and reduce the gap between synthetic and real data. For robustness purposes, several pre-processing steps are applied to the interferograms used in [1] to include atmospheric phase, snow cover, or water vapor in the estimation of the deformations. The proposed CNN is then used for the early detection of the eruption of Turkish volcanoes with Sentinel-1 interferograms. The SimCLR framework is leveraged in [32] to train a CNN feature extractor unsupervisely and to detect volcanic deformations from a small number of annotated samples. Nevertheless, knowledge acquired from synthetic data is hardly transferrable to real datasets, and often results in a large amount of false positives [35]. To limit this issue, Bountos et al. [36] propose a novel manually annotated dataset: Hephaestus. The latter is dedicated to deep learning training for the detection of unrest periods of 44 of the world's most active volcanoes. It is composed of 19919 Sentinel-1 interferograms, among which 1833 show actual volcanic activity.

Similar to the volcano domain, deep learning is increasingly being applied to InSAR data to detect slope instabilities such as landslides and rock glaciers. It is worth mentioning that InSAR provides critical insight on the state of activity, the velocity, and the threat of these phenomena. Bertone et al. [10] update 11 inventories of rock glaciers by including an estimate of their velocity based on InSAR images. Interferograms are used to detect new locations of slope instability in [11], where the authors use Sentinel-1 interferograms to map moving areas in the northern Norway. Bodin et al. [12] create a new inventory of rock glaciers over the French Alps by coupling InSAR and optical data. Standard deep learning architectures have already been trained to detect slow-moving landslides on InSAR products. Mao et al. [37] suggest improving YOLOv8, while Chen et al. [38] combine YOLOv5 with a Swin Transformer to complete this task. To enrich and encourage further deep learning studies aimed at slope instability detection on interferograms, we recently proposed the ISSLIDE dataset [13]. The latter is manually annotated to generate 13230 patches of interferograms surrounding 200 moving areas and adapted for deep learning approaches, as suggested in the baseline experiments. Zhang et al. [39] explore the dataset by encoding the interferogram in a multibranch architecture (MB-Net), which exploits independently the phase, sine, and cosine of the interferograms. Landslide monitoring is also targeted in [40], where a deep network is trained to compute an atmospheric delay map based on the digital elevation model (DEM) to reduce false positives. Burrows et al. [3] use the coherence map of the interferograms to segment sudden landslides. Finally, Guo et al. [41] exploit the temporal dimension by using InSAR time series together with hydrological records as input to the network to improve the detection of landslide deformations.

Such a variety of applications illustrates the great potential of using interferograms for slope instability monitoring. One of the reasons that limits the use of deep learning approaches in this context is their poor understandability. However, end users expect a clear and reliable justification of the detection before taking any risk mitigation action. Few approaches that target ground deformation detection on interferograms propose to evaluate their performance not only quantitatively but also in terms of explainability. Brengman and Barnhart [15] propose a deep CNN, which classifies whether synthetic interferometric patterns are due to the presence of ground deformations or to atmospheric and topographic effects. A synthetic dataset is also used in [16] to train a novel U-Net-like architecture based on separable convolutions [42] and attention mechanisms [43]. In any case, the authors proposed a visual explanation method

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based on CAMs [17], [18] to justify the performance and the reliability of the proposed algorithms. Beker et al. [44] exploit CAM to identify the difficulties of three CNN architectures when detecting volcanic deformation on InSAR products.

B. Explaining With Class Activation Maps

Among the growing number of neural network explainability strategies, CAMs have been especially used because of their high comprehensibility, straightforward computation, and potential for downstream tasks. In particular, the original CAM proposed in [17] consists of a linear combination of the feature maps of the last convolutional layer of the network. A global average pooling (GAP) layer is needed to compute the linear weights of the combination. This CAM is the starting point for numerous derived CAMs.

1) Gradient-Based CAMs: To propose the computation of a CAM at different layers in the network, the authors of Grad-CAM [18] suggest to use the average of the back-propagated gradients of each map as weights for the linear combination. Similarly, Grad-CAM++ [45] is an extension of Grad-CAM, which improves the saliency of the smallest features. Rebuffi et al. [21] point out that both of them lack detailed consistency since the gradient map is averaged to compute the linear weights. They propose to multiply each feature map element-wise with its corresponding gradient map and apply a Frobenius norm to merge them into NormGrad. The same strategy is proposed in [22] by replacing the Frobenius norm with a sum to generate Layer-CAM.

2) Mask-Based CAMs: Another approach for CAM generation consists of evaluating the performance of the network when masking the salient region or the background region of the input image. Such operations expect a strong decrease or increase in classification score, respectively. Ablation-CAM [19] weights each feature map based on the loss in class score when the given map is removed. Score-CAM [23] upscales the features to mask the input image, keeping only the regions highlighted by the given features. The masked images are then passed to the network to evaluate the benefit in class score once the background is removed. Naidu et al. [46] propose a variant of Score-CAM by progressively masking the input image with the feature map. The so-called IS-CAM is more computationally expensive but allows a more precise evaluation of the importance of each feature map. Wang et al. [47] propose the Feature Similarity Group-CAM (FSG-CAM) to reduce the computational cost of mask-based CAMs by grouping features based on their similarity before masking the input image. Finally, the CAM proposed in [48] is a negative-impact extension of Score-CAM, where the input image is masked twice: once to keep only the regions highlighted by the features, and once to keep the background. Therefore, both the gain and loss of score are used to weight the importance of the feature maps in the final decision.

3) Complementarity CAMs: These approaches propose to improve the quality of CAMs in less salient regions to enhance their localization capabilities. In particular, the CAM generated by the ACoL strategy [20] is obtained by combining the CAM of two parallel classifiers. The CAMs of both classifiers are complementary: the input of the second classifier is masked with the CAM of the first one. Therefore, the second is forced to find new features to accurately classify the images. Similarly, ReCAM [49] leverages two distinct classifiers. Its strategy is to train the first one to multilabel the images, generate a CAM for each of the classes, and use them to mask the input image. The second classifier is then trained to mono-label each masked image. This forces the network to find more discriminative features, which helps in generating a more detailed CAM.

4) More CAMs: CAM generation is still an ongoing research area. Among the recent methods, Wang and Bunyak [50] propose DFT-CAM and use the Discrete Fourier Transform to find the most relevant features in a given layer. Eigen-CAM is computed in [51] by computing the eigen vectors of the activation maps of each target layer. To find more discriminating features and not only the salient region of the input image, Chen and Sun [52] implemented LPCAM. It selects prototypes of features for each class and computes the cosine similarity between the image and each prototype of the class.

5) Attention-Based CAMs: The last category of CAMs is more related to attention modules than to model explanation strategies. Zhang et al. [53] propose to learn the weights to be used to linearly combine the feature maps and generate the most accurate CAM during the training process. The LFI-CAM proposed in [54] uses an auxiliary "feature importance network" to generate the attention map and explain the behavior of the main network. Zhu et al. [55] propose to generate both the target and background CAMs based on a "mutually-exclusive aggregator" that behaves similarly to an attention module.

C. Using Class Activation Maps

Based on Section III-B, a wide variety of CAMs can be implemented to understand the decisions made by the networks a posteriori of their training. The standard downstream task applied to these CAMs consists of object localization—e.g., for detecting cracks in buildings with a standard CNN architecture, as in [30]. Some approaches even use them for segmentation purposes, as in [5], where Grad-CAM is used to create a segmentation map of damaged buildings after an earthquake event.

CAMs are also widely explored for weakly supervised learning. They serve as pseudo-labels to train a network to segment the classes of interest. The UM-CAM framework [56] consists of training a classifier to detect fetal brains from medical images of pregnant women. A multilayer Grad-CAM is computed to serve as a pseudo-annotation map for the training of an auxiliary segmentation network. Wang et al. [57] also propose using CAM-derived pseudo-labels to guide an automatic medical reporting network. The same strategy is explored in [58] by refining the CAM with conditional random field (CRF) for standard object segmentation. CAMs are used as pseudo-labels in [59] to initialize a level set approach for semantic segmentation. Such weakly supervised approaches have been widely extended to remote sensing images to address the lack of annotations. Liu and Zhang [60] use the CAM computed from multiple layers of a classifier to

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generate pseudo-labels for a dual-stream segmentation branch. The latter aims at localizing urban areas from optical images. An interactive loss is introduced to make the training of both networks mutually beneficial. Grad-CAM is leveraged in [61] to generate pseudo-label maps from a teacher network, which are used to train a student network. The same CAM is used in [62] to increase the training dataset size of a segmentation network based on feedback blocks. A curriculum training is applied by adding noise to the pseudo-labels and removing it progressively during the training. Curriculum learning is also applied in [63] by progressively increasing the uncertainty threshold used to binarize the pseudo-labeled annotation maps. The latter are obtained by generating a CAM similar to the ACoL algorithm.

CAMs can also be used to strengthen the training of neural networks. Qin et al. [64] propose to train a two-branch neural network and compare their respective CAM to localize objects in an image. Each branch is designed to capture different and complementary salient features. The comparison of CAMs is also explored with Grad-CAM++ in [45] to transfer the knowledge from a teacher to a student network. The SEAM neural network [65] based on attention mechanisms is trained to include more details in the original CAM [17]. Morales et al. [66] propose to use the CAM during the training by randomly masking the salient part of the input images with the CAM. Thus, the network is forced to learn new discriminative features with a lower saliency. Finally, the Puzzle-CAM in [67] splits the input image into non-overlapping tiles and computes their CAM. Training the network to be consistent between the CAM of the whole image and the CAM reconstructed from the tiles helps to highlight more variable features.

Although all previous methods propose to use the CAM to train neural networks, it does not make them more explainable. Only a limited number of approaches have been proposed to constrain their explainability and ensure their reliability. In particular, the Right for the Right Reasons strategy [68] is one of the first approaches that calculates the training loss based on the explanation performance. It compares the gradient backpropagated to the input image with an expected annotation map. Similarly, Erion et al. [69] propose to constrain the network explanation to be consistent from one pixel to another by using the expected gradients method. Schöttl [70] suggest training a classifier with three new CAM-based loss terms: the CAM entropy, the CAM ellipsoidal area, and the CAM dispersion. Similarly, Crum et al. [71] propose five new metrics to evaluate the quality of the class activation maps based on their entropy and robustness to disturbances (e.g., noise and geometric transformations). Due to the lack of pixel annotations, Wargnier-Dauchelle [72] propose to partially constrain the CAM to detect pathological structures of the brain in medical images. The training loss forces the CAM of non-pathological image samples to be empty. Pathological ones are not constrained on their CAM because no ground truth is available. The authors of CYBORG [73] train a network to discriminate between real and GAN-generated images. To this end, the CAM of the network is constrained to match human-like saliency maps. The latter are annotations manually generated by asking a panel of participants to discriminate the images and to highlight the regions that support their decisions.

As shown by the above review, a very wide range of approaches propose to explain the behavior of a network a posteriori, but very few of them force the network to be consistent when explaining. Nevertheless, in the context of slope instability detection, high expectations are set on understanding the decisions of deep neural networks. Therefore, training a network to explain itself is of crucial importance in this field to better understand, prevent, and monitor the threats. To the best of our knowledge, our ECSPLAIN framework is the first to ensure classifier explainability for remote sensing images and, in particular, for the detection of moving areas on SAR interferograms.

IV. Method

A. Motivations

The proposed framework is implemented to meet the multiple expectations of the different actors involved in slope instability studies.

1) Geomorphologists and InSAR Experts: In this article, the proposed method aims to follow a geomorphological routine for detecting moving areas. Geomorphologists rely on specific geomorphological attributes to detect and monitor moving areas in InSAR data [13] (e.g., bubble and fringe patterns, continuity in time, consistency across temporal baselines). By training a network to explain itself using these attributes, we encourage non-InSAR experts to exploit this modality while being consistent with their expectations. In this work, we focus on generating a localization map to justify a decision. Therefore, our framework produces two outputs: a binary classification to detect the presence or absence of moving area in the image, and a localization map to highlight the regions affected by the phenomenon. This brings three major advantages. First, when a moving area is detected, the network produces a localization map that more accurately reflects the uncertainty of the movements near the boundaries of the instability. Second, when a moving area is missed, geomorphology experts can still produce the localization map to ensure the relevance of the detection or identify suspicious areas despite the non-detection. This ensures that the end user keeps control over the final decision, unlike a segmentation network, which generates a zero map in case of a misdetection. Third, when a moving area is wrongly detected, InSAR experts can easily identify counterfactual patterns. InSAR experts can then update the framework by including more physical constraints in the training process to reduce the detection of these false positives and focus the network on the actual patterns of interest.

2) Deep Learning Experts: From a computer science perspective, segmenting from a classifier brings significant benefits, as illustrated in Table I. First, standard segmentation networks require a greater number of parameters than classifiers (e.g., >25M for ResNet50 [74] versus >42M for DeepLabv3 [76] with a ResNet50 backbone). Therefore, these networks are more prone to overfitting on small datasets. In addition, the decoder increases the number of floating point

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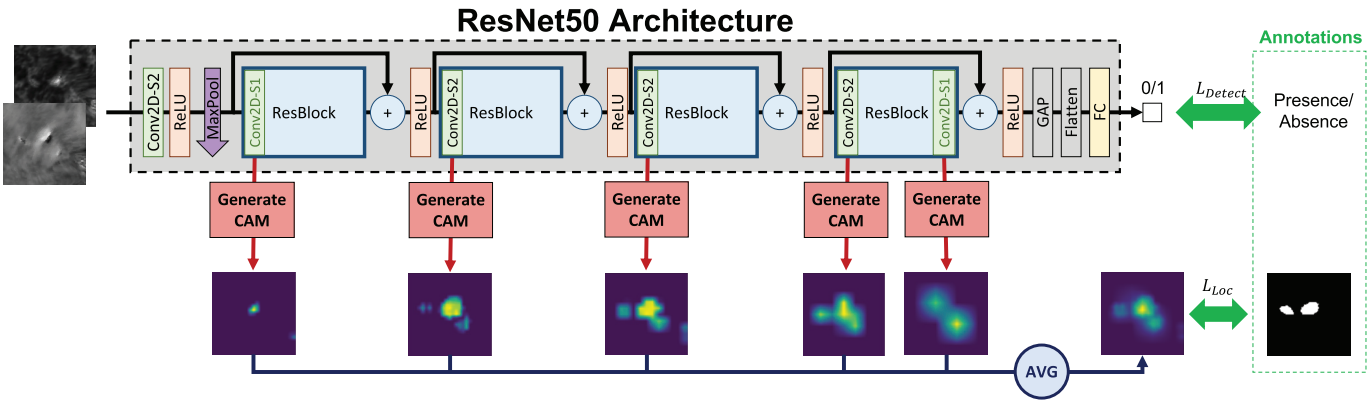


Fig. 1. Scheme of the ECSPLAIN framework. The real and imaginary parts of the interferograms are classified by the ResNet architecture to compute the classification loss L_{Detect}. The CAM is computed for multiple layers to explain the decision. The combination of these CAMs is compared to the actual annotations to compute the segmentation loss L_{Loc}. In the scheme, the stride of the convolution is specified by i in Conv2D-Si. The circled AVG refers to the CAM aggregation given in (3).

TABLE I
INTERNAL CHARACTERISTICS OF CLASSIFIERS AND SEGMENTATION NETWORKS AS IMPLEMENTED IN PYTORCH

	#Params (M)	FLOP (G)	Mem. Req. (MB)
Classifier Networks			
ResNet18 [74]	11	1.81	44
ResNet50 [74]	25	4.09	97
Segmentation Networks			
FCN [75]	35	152.72	135
DeepLabV3 [76]	42	178.72	160

operations (FLOPs)—from 4.09G for ResNet50 to 178.72G for DeepLabV3. As a result, classifier approaches are faster to compute, consume less power, and require less memory (97 MB for ResNet50 against 160.5 MB for DeepLabV3) than the traditional segmentation networks.

In conclusion, the proposed approach leverages explainability to make a classifier able to justify its decisions by outputting the localization map of the moving areas. This strategy not only satisfies the need for understanding in geosciences but also ensures the development of lightweight and effective deep learning approaches.

B. ECSPLAIN Training Strategy

Based on the aforementioned literature and motivations, in this article, we propose to introduce a novel strategy to segment moving areas through classification explainability. In particular, we aim to train an ECSPLAIN. This framework constrains the CAM of a classifier during the training phase to match the polygon of the moving area provided by the ground truth. This constraint mimics the first step of the routine defined in [13]. To the best of our knowledge, this is the first framework that leverages explainability at training time to detect natural phenomena in remote sensing data.

The forward pass of this novel training strategy depicted in Fig. 1 is four-stepped as follows.

1) *Detecting moving areas:* The input batch is passed forward to the selected classifier network to predict the presence of a moving area on each interferogram (black arrows in Fig. 1). The decisions are then compared to the true labels to compute the detection loss (L_{Detect}).
2) *Explaining the decisions:* Following the decision taken by the classifier, a CAM is computed at different target layers T, as illustrated by the red arrows in Fig. 1. The computation of the CAM at different layers aims to improve the detection of smaller and less salient regions as experienced in [21], [22], [56], and [60].
3) *Generating a multilayer CAM:* This step is materialized by the blue arrows in Fig. 1. The different CAMs computed at the previous step are averaged to compute a multilayer CAM, as given in (3). Following the methodology suggested in [18], [21], [22], and [23], each CAM_L computed at layer $L \in T$ is refined by keeping the regions with a positive score, normalizing them from 0 to 1 and upsampling them to the original image size with a bilinear interpolation

$$\text{CAM} = \frac{1}{\#T} \sum_{L \in T} \text{Up}\left(\frac{\text{ReLU}\left(\text{CAM}_L\right) - \min_L}{\max_L - \min_L} \right) \quad (3)$$

where ReLU and Up stand for Rectified Linear Unit and Upsampling, respectively. The notations $\min_L$ and $\max_L$ are the minimum and maximum values of the $\text{ReLU}(\text{CAM}_\mathbf{L})$ generated in the layer L. $\#T$ is the number of targeted layers.
4) *Localizing moving areas:* The final CAM resulting from (3) is considered as the localization map generated by the classifier network. The key idea of the ECSPLAIN framework lies in an additional loss term (L_{Loc}), which forces the network to match this localization map with the actual annotation map. If no moving area is present on the input interferogram, the CAM is matched to a zero map. This ensures that the decisions are based on human-like reasoning, i.e., by localizing accurately the regions which support the detections.

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In this work, several choices have been made to illustrate the efficiency of the novel training strategy. However, these choices are not intended to be optimal. The objective of this article is to focus on the benefits of the training strategy, not on the impact of the hyperparameterization. Thus, all of the following choices can be optimized further to improve the performance obtained in Section VI.

Among the possible classifier architectures, ResNet-derived approaches [74] have demonstrated significant performance and applicability in the literature [26], [44], [47], [71], [77]. These networks use residual connections to speed up the training, limit the vanishing gradient issue, and maintain low-level feature availability throughout the network. In addition, the global average pooling (GAP) layer in the final stages makes it suitable for variable input shapes. The architecture is also widely used as a backbone encoder for segmentation networks, including FCN [75] and DeepLabV3 [76], which are used as comparative methods in Section VI. Therefore, ResNet50 is selected as the backbone architecture for the ECSPLAIN framework. Since this architecture is composed of different stages, the T targeted layers, which are used for the multilayer CAM computation, are selected at the beginning of each stage. In particular, the four post-pooling convolutional layers are chosen because the pooling operation filters the most relevant features—i.e., the most helpful for the decision. The very last convolutional layer is also selected as it is the most semantically informative layer. This work implements and compares four CAMs: Grad-CAM [18], Layer-CAM [22], NormGrad [21], and Score-CAM [23]. Section IV-C details the computation of these CAMs. Finally, (4) gives the loss function used to train the network. It is composed of the detection L_{Detect} and the localization L_{Loc} errors, which are based on the binary cross-entropy loss. The latter is applied pixel-wise between the multilayer CAM and the annotation map to compute L_{Loc}

$$L\left(\left[\hat{y}, \text{CAM}\right], \left[y, A\right]\right) = L_{\text{Detect}}\left(\hat{y}, y\right) + \lambda L_{\text{Loc}}\left(\text{CAM}, A\right) \quad (4)$$

with λ a hyperparameter used to moderate the importance of the explanation compared to the detection. $\hat{y}$ and y are the network prediction and the ground-truth label, respectively. A is the annotation map localizing the target moving area.

C. Selected CAMs

The standard algorithm for CAM generation consists of a linear combination of the different feature maps. For the sake of readability, in this section, matrices are displayed in **bold**, while scalar values are left unchanged. Thus, by considering F_L^k as the kth feature map of a given layer $L \in T$ with N_L channels and α_L^k as the linear weights used for the combination, the heat map CAM_L is generated as described in the following equation:

$$\text{CAM}_L = \sum_{k=1}^{N_L} \alpha_L^k \odot F_L^k. \quad (5)$$

The most challenging task is to find the appropriate weights α_L^k used for the combination. Depending on the method used to generate the CAM, the $\odot$ symbol can be a multiplication by a scalar α_L^k or the Hadamard product between the feature map and a weighting matrix α_L^k. In the following, we consider L to be fixed for describing the CAM (i.e., $F_L^k = F^k$). Among the wide variety of CAM methods discussed in Section III-B, four of them are explored in this article.

1) Grad-CAM [18]: Together with the original CAM [17], the Grad-CAM is one of the most used CAMs for the explanation task. It consists of backpropagating the gradient of the output score of a given class to the target layer. The authors argue that a positive gradient reflects high importance features. They propose to combine the feature maps by weighting them with the average of their gradient map as in the following equation:

$$\alpha_{\text{GC}}^k = \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} \frac{\partial \hat{y}}{\partial {F^k}_{ij}} \quad (6)$$

with F_{ij}^k the pixel of the kth feature map at location i, j, and $\hat{y}$ the score of the class of interest output by the network. H and W are the dimensions of the feature maps.

The major advantage of this approach over the original CAM is its applicability to any layer of the network without requiring architectural changes or retraining.

2) Layer-CAM [22]: Similar to the Grad-CAM, the Layer-CAM uses the gradient to compute the weights α^k. But, instead of averaging them, the authors propose a local strategy. The gradient maps are first thresholded to keep only positive gradients, as shown in (7). Then, each gradient map is combined with its corresponding feature map using the Hadamard product $\odot$ in (5)

$$\alpha_{\text{LC}}^k = \text{ReLU}\left(\frac{\partial \hat{y}}{\partial F^k}\right) \quad (7)$$

with ReLU(.) corresponding to the Rectified Linear Unit, and α_{LC}^k the weight map of size $H \times W$.

3) NormGrad [21]: The main difference between Layer-CAM and NormGrad is the aggregation function used to merge the feature maps. Instead of using a standard linear combination, the authors propose using the Frobenius norm to compute the CAM at any layer of the network. Therefore, (5) switches to the following equation:

$$\textbf{NormGrad} = \left|\left|\alpha_{\text{LC}}^k \odot F^k\right|\right|_F \quad (8)$$

where $\|.\|_F$ is the Frobenius norm and $\odot$ is the Hadamard product. The same weights α_{LC}^k as for Layer-CAM are used to emphasize more local features.

4) Score-CAM [23]: Unlike the previous three maps, Score-CAM uses a masking approach to avoid vanishing gradient issues. The linear weights α^k are computed by evaluating the score increase for the target class when only the features highlighted by each feature map are input. Fig. 2 illustrates the process. First, the input is passed through the network, as illustrated in block 1. Then, in block 2, each feature map is upsampled to the input image resolution and normalized from 0 to 1. The input is then masked by each feature map to remove non-salient regions before being fed back into the network. The relevance of each feature map is then calculated

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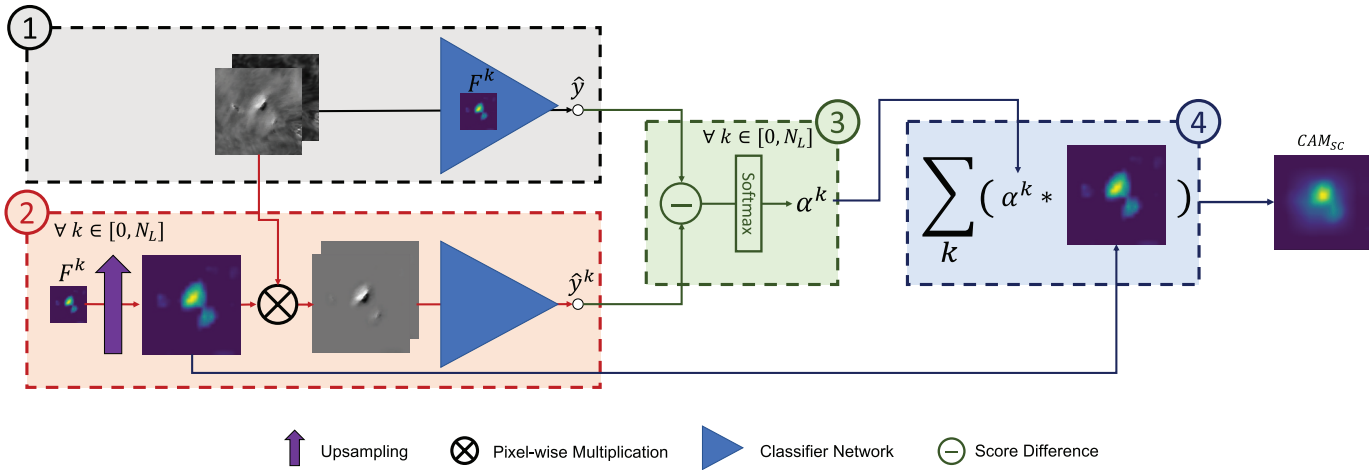


Fig. 2. Score-CAM generation framework. The input image is first classified. Each target feature of the target layer is upsampled to map the original input. Each masked image is then classified by the network to figure out the evolution of the class score. The final CAM is computed by combining the feature maps with its corresponding score evolution.

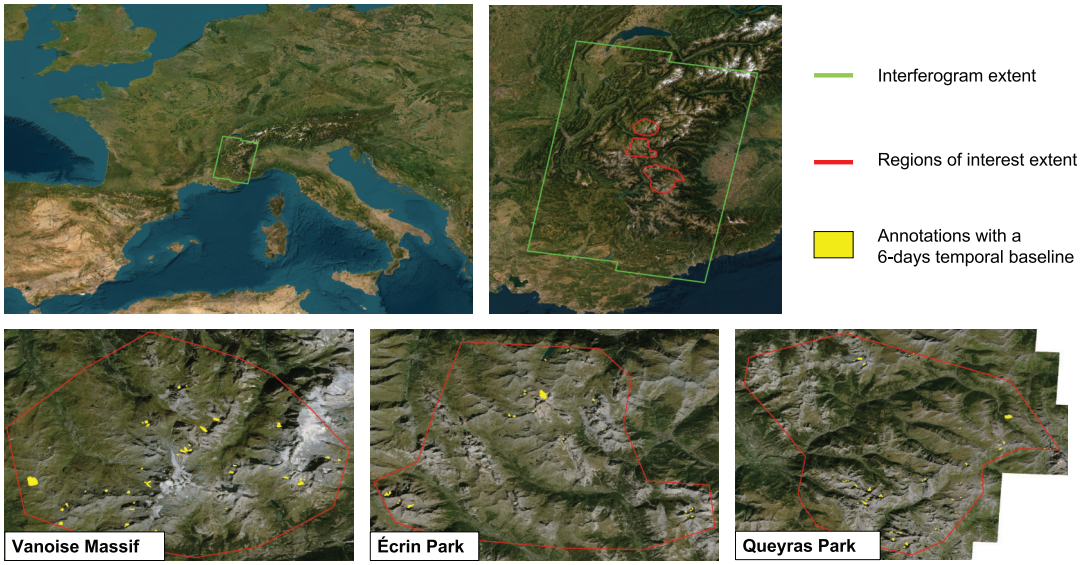


Fig. 3. Geographical location of the regions of interest for the current work. The top row illustrates the area covered by the studied interferograms. The bottom row illustrates the available annotations obtained on the interferograms with a temporal baseline of 6 days for the three regions of interest.

in block 3 based on the score obtained when the masked image is input to the network, as given in the following equation:

$$\alpha_{\mathrm{SC}}^{k} = \mathrm{Softmax}\left(\hat{y}^{k} - \hat{y}\right) \tag{9}$$

where $\hat{y}$ and $\hat{y}^k$ are the scores of the given class for the input image and the image masked with the kth feature map, respectively. The Softmax activation function is used to ensure positive weights. The final block 4 illustrates the linear combination of the feature maps given in (5).

V. Experimental Setup

A. Dataset

The experiments are performed on a dataset which we recently created. The ISSLIDE dataset [13] is dedicated to the localization of slow-moving areas on wrapped interferograms and available on IEEE Dataport (https://ieee-dataport.org/documents/isslide-insar-dataset-slow-sliding-area-detection-machine-learning). The latter are based on 19 Sentinel-1 SAR images spanning from July to October 2018 over the French Alps. They cover more than 49 500 km^2, as seen in the top images of Fig. 3. The 54 orthorectified interferograms are computed with a temporal baseline of 6, 12, and 18 days. Each interferogram is composed of two different swaths, each 284 km long and 85 km wide.

The annotation process focused on the identification of rock glaciers and slow-moving landslides in three regions delineated in red in Fig. 3: the Vanoise Massif, the Écrin Park, and the Queyras Park. Additional movements have also been found outside of these areas, for a total of 200 identified moving areas. Based on manual annotations, the interferograms are

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further processed to deliver a dataset of more than 13 000 interferogram patches surrounding the identified instabilities. Among them, 8701 patches show actual motion patterns. The delivered patches consist of 100×100 pixels coherence (ρ) and phase difference (φ) maps. They are assigned to the annotation map of the corresponding moving area.

Due to the mono-interferogram approach conducted in this study, only the 8701 patches are retained for the training. In addition, a total of 4740 patches are randomly extracted from the raw interferograms to serve as background-only samples for the classification task. These patches are filtered to ensure no overlap with the annotated patches.

To ensure that the training and evaluation sets do not overlap, the same method as in [13] is applied for the splitting: all the patches within the Queyras Park are kept for the evaluation dataset, while the others belong to the training set. In addition, only the patches with actual ground motions extracted from 80% of the interferograms are included in the training set. Those over the Queyras Park extracted from the remaining 20% of interferograms are used for the evaluation step. The background-only images are only split depending on the interferogram they belong to. Therefore, the final training and evaluation datasets contain 8,483 (4817 patches around moving areas versus 3676 background-only patches) and 1633 (569 versus 1064) patches, respectively. The same data augmentation strategy as in [13] is used. It consists of adding a random offset to the phase difference map at each training epoch to improve the robustness to phase rotation. To avoid complex values as input to the network, the interferogram is decomposed into real and imaginary parts $[\rho\cos(\varphi), \rho\sin(\varphi)]$. In this way, the coherence is appropriately exploited to weight the reliability of the phase difference map prior to the feature extraction process.

B. Training Parameters

The ECSPLAIN approach is implemented with the ResNet50 architecture under the PyTorch implementation. The first and last layers are adapted to take the two-channel images as input and output a single bin for the detection. In addition, all batch normalization layers are removed first to avoid computational errors during the training due to CAM calculation, and second, because we used a batch size of 8, which is too small to extract representative statistics. All networks are trained from scratch for 194 epochs, which allows fair comparisons between the CAMs and limits the overfitting observed in offline experiments. The classifiers are updated with an Adam optimizer with a learning rate of $1e^{-4}$. The value of λ in (4) is set to 1 to give the same importance to the classification and the localization tasks. The only difference from one configuration to another is the method used to generate the CAM in the ECSPLAIN framework. This aims to better evaluate their impact on the performance without external factors. To compare the ECSPLAIN strategy with the standard a posteriori CAM localization methods, the ResNet50 network is also trained with $\lambda = 0$ in (4) and with the same hyperparameters.

Due to memory issues, a feature dropout is used to compute Score-CAM during the training. A subset of $n_L = N_L/10$ maps is selected and used for the Score-CAM generation. The feature maps are selected randomly for each batch, avoiding their specialization for the segmentation task. At the inference stage, all feature maps are used to generate the Score-CAM localization map.

C. Evaluation

The performance of the ECSPLAIN framework is compared to standard segmentation architectures proposed as baselines in [13]: FCN [75], DeepLabv3 [76], and an encoder-only residual version of U-Net [78], hereafter called ResUNet. The segmentation networks are trained to minimize the binary cross-entropy loss pixel-wise on the generated segmentation maps. The same hyperparameter configuration as detailed in Section V is used except for the learning rate and the number of training epochs. To avoid overfitting due to the number of parameters in these networks, FCN and DeepLabv3 are trained with a learning rate of $1e^{-7}$ for 1000 epochs. Since the ResUNet architecture is lighter, the learning rate is set to $1e^{-5}$ for 1000 epochs.

All methods are evaluated with classification and segmentation metrics.

1) Classification Metrics: These metrics are only applied to the detection task of the networks. Overall Accuracy (Acc.), Precision (Prec.), Recall (Rec.), and $F1$-score ($F1$) are used for the evaluation.

2) Segmentation Metrics: To evaluate the performance in the localization task, the Dice score and the intersection over union (IoU) are used as ensemble metrics. The Hausdorff distance (Haus. Dist.) is employed to evaluate the proximity of the predicted map to the true map. The Area Under the receiver operating characteristic Curve (AUC) is used to evaluate the network independently of the threshold used for the binarization. Finally, due to the uncertainty near the boundaries of the moving areas, the mean absolute error between the predicted probability and the blurred annotation is computed.

Classification metrics are computed by thresholding the output of the classifiers by 0.5. Those of the segmentation-only networks are calculated by assuming a detection once the generated segmentation map identifies 25 pixels with a score greater than 0.5. The segmentation metrics are computed to maximize the localization ability of the networks. Therefore, for each method, the threshold used to compute the Dice, the IoU, and the Hausdorff distance is selected to maximize the Dice score.

VI. Results

Based on the experimental setup, this section aims to evaluate the ECSPLAIN framework from two perspectives. First, its segmentation performance is evaluated in comparison to standard segmentation networks from the literature. The goal is to evaluate whether the framework achieves competitive performance with these non-explainable heavy networks. Second, its detection and localization performance are evaluated in comparison to the standard a posteriori explanation, i.e., a ResNet50 trained without guidance. In addition to comparing them on their explainability performance, we evaluate the

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TABLE II

QUANTITATIVE COMPARISON OF THE SEGMENTATION NETWORKS, A POSTERIORI CAM LOCALIZATIONS, AND ECSPLAIN FRAMEWORK. BOTH CLASSIFICATION AND SEGMENTATION METRICS ARE PROVIDED. THE LAST COLUMN INDICATES THE THRESHOLD USED TO BINARIZE THE LOCALIZATION MAPS, I.E., WHICH MAXIMIZES THE DICE SCORE. BEST AND SECOND-BEST RESULTS ARE BOLDED AND UNDERLINED, RESPECTIVELY

	Classification Metrics				Segmentation Metrics					Thresh.
	Acc. ↑	Prec. ↑	Rec. ↑	F1 ↑	MAE ↓	Dice ↑	IoU ↑	Haus. Dist. ↓	AUC ↑	
Segmentation Networks										
FCN [75]	95.83%	**96.30%**	91.56%	0.939	**1.36e-2**	0.661	0.523	25.52	0.953	0.09
DeepLabv3 [76]	93.02%	86.99%	94.02%	0.904	1.64e-2	0.622	0.480	36.62	0.955	0.12
Res. U-Net [74]	**95.96%**	93.89%	**94.55%**	**0.942**	1.43e-2	**0.684**	**0.547**	25.69	**0.964**	0.1
A posteriori explanation - ResNet50										
Grad-CAM [18]	94.67%	92.43%	92.27%	0.923	13.74e-2	0.161	0.093	58.01	0.524	0.10
Layer-CAM [22]					13.00e2	0.350	0.230	45.95	0.813	0.26
NormGrad [21]					13.76e-2	0.379	0.254	44.01	0.820	0.27
Score-CAM [23]					14.20e-2	0.360	0.244	40.82	0.858	0.37
ECSPLAIN - ResNet50										
Grad-CAM	95.04%	92.66%	93.15%	0.929	3.35e-2	0.637	0.504	25.54	0.949	0.3
Layer-CAM	94.00%	94.02%	88.40%	0.911	4.72e-2	0.570	0.435	26.45	0.920	0.28
NormGrad	95.89%	95.63%	92.44%	0.940	4.73e-2	0.575	0.439	25.98	0.923	0.3
Score-CAM	95.71%	94.47%	93.16%	0.938	5.92e-2	0.643	0.500	**24.08**	0.959	0.33

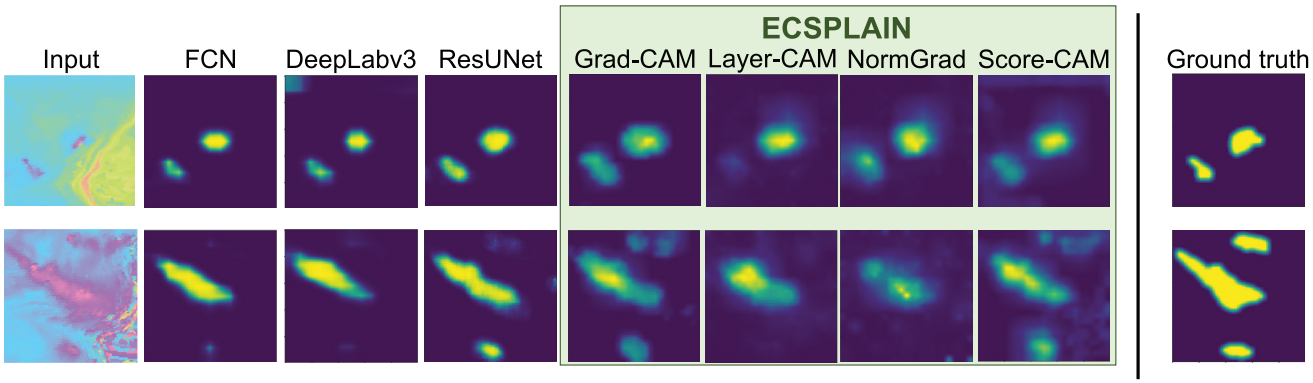


Fig. 4. Qualitative results comparing the performance of segmentation networks and the ECSPLAIN framework trained with different CAMs on evaluation images. Only the phase difference of the input interferogram is displayed for better visualization. All maps are ranging from 0 (dark blue) to 1 (yellow).

impact of the ECSPLAIN framework on the classification metrics. The quantitative results for both comparisons are available in Table II.

A. Segmentation Versus Localization

As expected, the segmentation networks achieve the best results due to the larger number of their parameters and their pixel-wise decision. Conversely, the ECSPLAIN strategy is based on a classifier that focuses on taking an image-wise decision-i.e., with a wider receptive field.

Nevertheless, the performance achieved by the proposed framework is highly competitive with that of the segmentation networks. In terms of classification metrics, ECSPLAIN trained with NormGrad and Score-CAM are the second and the fourth best moving area detectors overall. In terms of segmentation metrics, the frameworks trained with Grad-CAM and Score-CAM rank third and fourth, respectively, in Dice and IoU. Score-CAM achieves the best Hausdorff distance and the second-best AUC of all. Such performance can be better understood coupled with the qualitative results of Fig. 4. Based on the first three columns, the segmentation networks under-segment the actual localization of moving areas. In particular, when several instabilities are located within the patch (e.g., in the bottom row), their segmentation map focuses on the largest one. Conversely, ECSPLAIN, trained with Grad-CAM and Score-CAM, mostly focuses on highlighting each suspicious region. Such a behavior explains that the ensemble metrics are more accurate with the segmentation networks, but that the Hausdorff distance is greatly reduced with ECSPLAIN. In addition, the MAE score is four times better with the segmentation networks than with our framework. This is a consequence of the localization task targeted by the CAM generations. Since the segmentation task is binary, the decision is made without a margin of uncertainty. In the current ECSPLAIN framework, the detections are nuanced so that in the first row of Fig. 4, the localization map of the moving area is obtained with blurred boundaries.

These uncertainty maps better match experts' expectations for the mapping of slope instabilities. In addition, the early

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detection of moving areas is mainly used for prevention purposes. Therefore, the detection of false positives is preferred over the non-detection of true positives. Since segmentation networks are subject to undersegmentation or missing the small instabilities, they cannot be fully trusted for this task (only ResUNet can detect the small instability at the bottom of the second row in Fig. 4). The only approaches that succeed in detecting the presence of all moving areas in Fig. 4 are ECSPLAIN trained with the Grad-CAM or the Score-CAM. More details on the comparison between the CAMs are given in Section VII-A.

In conclusion, the ECSPLAIN framework not only solves the major drawbacks of segmentation networks mentioned in Section IV-A but its performance is also highly competitive.

B. Explainability or ECSPLAIN-Ability

Based on the segmentation performance in Table II, applying the CAMs a posteriori of a standard training leads to a strong lack of consistency in the localization of slope instabilities on interferograms. This is even more visible in Fig. 5, where all CAMs applied a posteriori of the training show poor localization ability despite obvious motion patterns. Conversely, when the CAM is generated in the context of the ECSPLAIN framework, the classification network succeeds in accurately localizing both the large slope instability and the smaller pattern at the bottom left of the image. Several factors can explain the counter-performance of the a posteriori CAM.

1) *The interferograms*: The nature of the input data is itself a challenge. Unlike standard images used for CAM generation, interferograms are complex-valued images. By separating the real and imaginary parts of the images, the objective was to mitigate the effect of this characteristic. However, due to the fact that the phase is rotational, the interferograms are wrapped, and few phase corrections have been applied, a given motion pattern can be extremely variable. Despite the fact that data augmentation is applied during the training by randomly shifting the phase, this task is still challenging for standard architectures without guidance. Further works may consider the use of complex-valued convolutional layers [79] to intend to alleviate the issue.
2) *The distortions*: Due to the mountainous nature of the study area, the SAR images—and therefore the interferograms—are strongly affected by geometric distortions [14]. In particular, the orthorectification of foreshortening and layover areas can produce patterns that mimic motion patterns. Distinguishing them is even more challenging because the actual moving area patterns are often close to these counterfactual patterns. Therefore, without supervision, the network may consider both the moving areas and the distortions as patterns of interest for the decision. The use of the segmentation map in the ECSPLAIN framework mitigates the impact of these counterfactuals by focusing the attention of the network on patterns related to moving areas.
3) *The saliency*: One of the major drawbacks of the standard CAM approach is its focus on the actually salient part of the image. As visible in the results of the Score-CAM applied a posteriori in Fig. 5, the most salient region is located within the main motion pattern. However, in the case of multiobject segmentation, the CAM may target the localization of the largest object and reduce the importance of less salient and smaller patterns. A possible solution would be to use complementary CAMs [20], [49], [64] to target not only the extraction of the most salient part of the image but also the related, less salient features.
4) *The selected layers*: Since the features are the core of the CAM generation, the selection of the layers of interest is crucial to ensure the accuracy of the generation. By generating the CAM with the layers that follow the pooling stages, the goal is to ensure that only the features with the highest activations are used to compute the CAM. Nevertheless, Rebuffi et al. [21] demonstrate that CAMs are sensitive to the depth of the layer used. In particular, they show that the Grad-CAM is reliable only when applied to the very last convolutional layer of the network but is inconsistent for the other layers. Based on the results of the Grad-CAM applied a posteriori both quantitatively and qualitatively, a similar conclusion can be drawn from our experiments.

Based on the results obtained with the ECSPLAIN framework, constraining the explanation of the network by using the segmentation map helps to guide the network to overcome all these issues and generate more accurate and insightful CAMs. In addition, guiding the network to justify its decision helps to improve the classification performance of the network. In Table II, the precision increased by more than three points and the recall increased by almost one point with ECSPLAIN compared to a classifier trained without guidance. This performance highlights that the guidance helps to reduce the false positive detections due to the counterfactual samples without negatively affecting the recall. Thus, the classifier can justify its decisions and is more robust against geometrical distortions.

VII. Discussion

This section targets a better understanding of the performance of the ECSPLAIN framework and the selection of the most appropriate CAM method.

A. CAMs Comparison

Based on the quantitative performance of the ECSPLAIN strategy in Table II and on the qualitative one in Figs. 4 and 5, the choice of using one CAM instead of another depends mainly on the application. The use of NormGrad for the training achieves the second-best results in terms of classification metrics and the first in the ECSPLAIN framework. Therefore, the NormGrad significantly helps the network to focus on the actual salient regions. But, from a segmentation point of view, NormGrad and Layer-CAM show a significant decrease in performance compared to Grad-CAM and Score-CAM. The main reasons for such a drop are the characteristics of the studied phenomena. The vast majority of slope instabilities show strong spatial correlations

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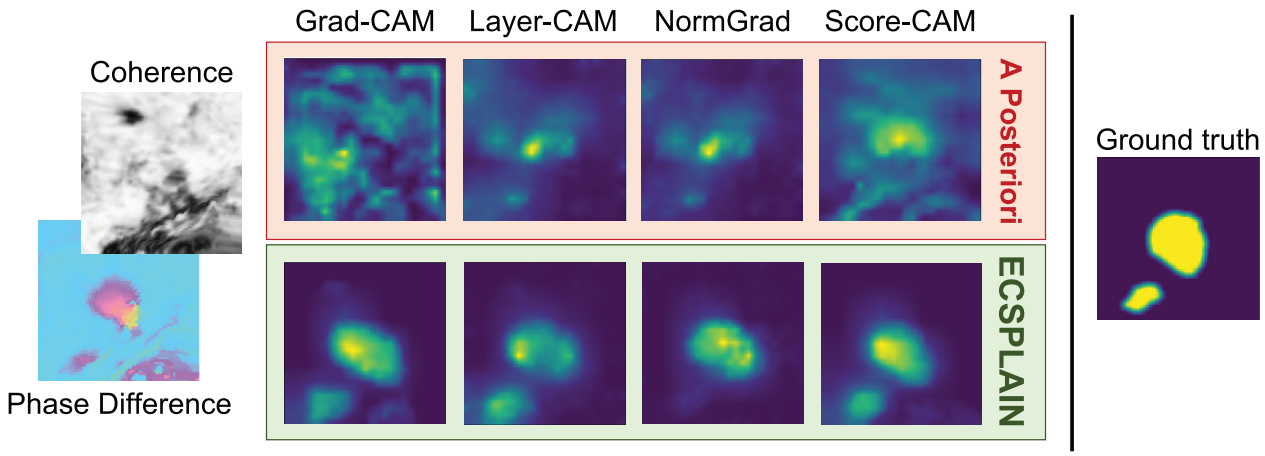


Fig. 5. Qualitative comparison between the a posteriori (top) CAM localization and the (bottom) ECSPLAIN framework. For visualization purposes, phase and coherence are displayed instead of the real and imaginary parts of the interferogram.

and very compact patterns. This favors the methods with a global receptive field and harms those with a local one—i.e., NormGrad and Layer-CAM, which implement a pixel-wise weighting strategy (see Section IV-C). Such operation can prevent the network from behaving similarly between two pixels in the same neighborhood. In addition, these two CAMs lack multiobject capabilities: Layer-CAM fails to identify the smaller motion pattern in the first row of Fig. 4 and NormGrad fails on the one in Fig. 5.

Conversely, the behavior of Grad-CAM and Score-CAM is more in line with standard expectations. From a quantitative point of view, they show the most competitive results compared to the segmentation networks. From a qualitative point of view, they are the only methods that can identify all the moving areas in Figs. 4 and 5. Among these two CAMs, Score-CAM achieves the best balance between classification and segmentation. However, one of the main computational advantages of the ECSPLAIN framework is its low requirements in terms of parameters and FLOPs. On the one hand, the computation of the Grad-CAM requires a single forward and backward pass to generate both the detection and the localization map. Thus, the number of FLOP is of the same order of magnitude as the original classifier network. On the other hand, computing Score-CAM requires the same number of forward passes as there are extracted feature maps in each targeted layer. Therefore, for a ResNet50 architecture with the five selected layers, the Score-CAM requires 1472 passes through the network. This is the reason why, to be consistent with the motivations detailed in Section IV-A, the ECSPLAIN framework coupled with the Grad-CAM is more interesting than its competitors in this work.

B. CAM Generalizability

This section proposes a deeper study of the generalizability of the explanation capabilities of the classifiers trained with the ECSPLAIN framework. In particular, Table III highlights the performance of the classifiers when explained by another CAM method at inference time. These results support the conclusion regarding Layer-CAM and NormGrad in the previous section: due to their similarity in generation, their performance is mostly not affected when the inference explanation switches from one to the other.

TABLE III

QUANTITATIVE COMPARISON OF THE INFERENCE EXPLANATION GENERALIZABILITY OF THE ECSPLAIN FRAMEWORK DEPENDING ON THE CAM OF TRAINING. ONLY SEGMENTATION METRICS ARE PROVIDED. BEST INFERENCE PERFORMANCE FOR EACH FRAMEWORK IS BOLDED. BLUE ITALIC FONTS REFER TO THE REFERENCE INFERENCE, WHERE THE SAME CAM IS USED FOR THE TRAINING AND THE INFERENCE

Inference CAM	MAE ↓	Dice ↑	IoU ↑	H. Dist. ↓	AUC ↑
ECSPLAIN with Grad-CAM					
Grad-CAM	*3.35e-2*	*0.637*	*0.504*	*25.54*	*0.949*
Layer-CAM	15.95e-2	0.494	0.354	35.38	0.850
NormGrad	16.40e-2	**0.519**	0.378	**30.24**	0.865
Score-CAM	**14.91e-2**	0.518	**0.382**	32.85	**0.920**
ECSPLAIN with Layer-CAM					
Grad-CAM	**4.15e-2**	0.342	0.225	51.38	0.817
Layer-CAM	*4.71e-2*	*0.570*	*0.435*	*26.45*	*0.920*
NormGrad	4.95e-2	**0.562**	**0.429**	**26.70**	**0.918**
Score-CAM	9.00e-2	0.477	0.339	31.65	0.911
ECSPLAIN with NormGrad					
Grad-CAM	**3.88e-2**	0.317	0.206	47.66	0.787
Layer-CAM	4.64e-2	**0.579**	**0.439**	**26.11**	0.924
NormGrad	*4.73e-2*	*0.575*	*0.439*	*25.98*	*0.923*
Score-CAM	8.78e-2	0.485	0.346	39.63	**0.925**
ECSPLAIN with Score-CAM					
Grad-CAM	**9.23e-2**	0.244	0.152	56.58	0.727
Layer-CAM	11.31e-2	**0.529**	**0.385**	40.83	**0.890**
NormGrad	12.24e-2	0.519	0.379	**35.00**	0.881
Score-CAM	*5.92e-2*	*0.643*	*0.500*	*24.08*	*0.959*

Regarding the other CAMs, the Score-CAM generated at inference time shows great capabilities in explaining the classifiers trained with another CAM. It achieves the best performance in explaining the classifier trained with Grad-CAM and stable performance for the others. Conversely, the Grad-CAM shows the worst metrics of all when applied to the

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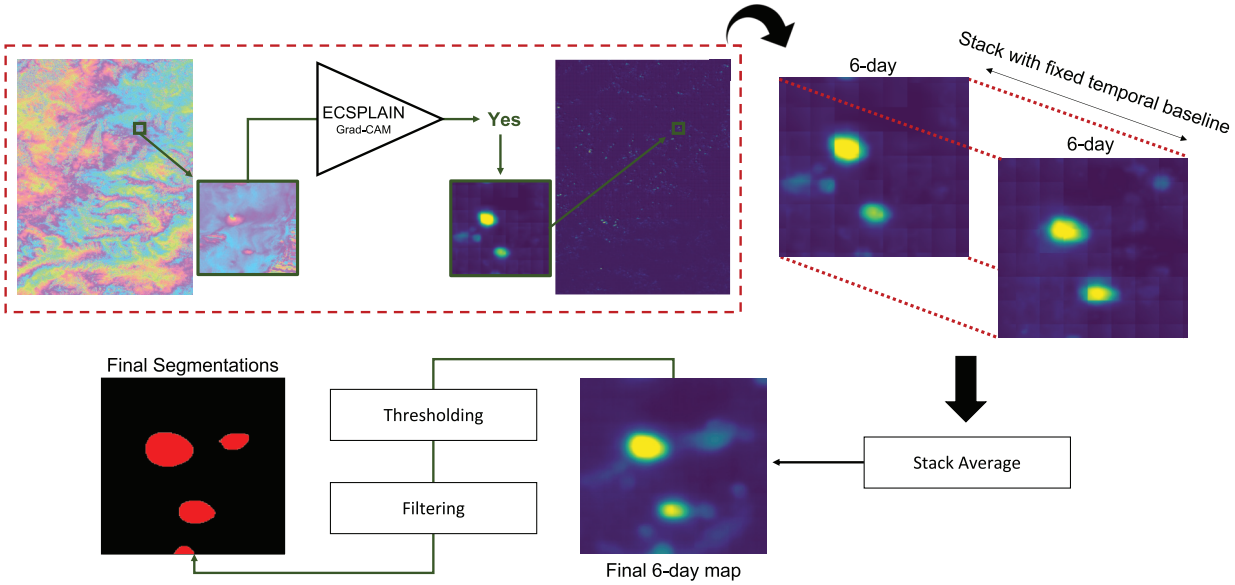


Fig. 6. General pipeline for the generation of the moving area localization maps and their detected polygons over the whole interferograms. For the sake of readability, the scheme illustrates only the case of interferograms with a six-day temporal baseline, and a small crop is used for visualizing the post-sliding window process, which is actually applied over the whole interferogram.

other classifiers at inference time. Two main reasons can be hypothesized: first, the Grad-CAM has been shown inefficient and inaccurate when computed in multiple layers in [21]. This behavior is only compensated with the ECSPLAIN framework because the Grad-CAM is constrained to produce accurate maps by using these layers, but its original behavior persists for the other classifiers. Second, all the other maps have been trained and constrained from a local point of view. Indeed, NormGrad and Layer-CAM use the Hadamard product in (5) and Score-CAM applies masking operations. But Grad-CAM is computed from a more global point of view by averaging the gradient maps. This limits its performance when applied to the classifiers trained with a local CAM.

From a generalizability point of view, all ECSPLAIN frameworks perform similarly when inferring with another CAM, except for Grad-CAM. Thus, by comparing these segmentation performance with that obtained in Table II by the CAMs applied a posteriori, the Dice and IoU scores are increased by more than 0.1 points. Similarly, the Hausdorff distance is decreased by up to 15 points, and the AUC gains up to 0.05 points. Therefore, the use of the ECSPLAIN framework not only achieves competitive results with segmentation networks when inferred with the trained CAM, but it also helps the classifiers to be more generalizable in terms of explainability strategy. For the sake of complete generalizability, ECSPLAIN trained with Grad-CAM is recommended over the others to ensure high performance for both global and local CAMs.

VIII. Enriching the Dataset

Based on the aforementioned conclusions, in this section, we propose to enrich the ISSLIDE dataset [13]. The ECSPLAIN framework is applied to detect more moving areas on the provided raw interferograms and to generate their localization map.

A. Detection Pipeline

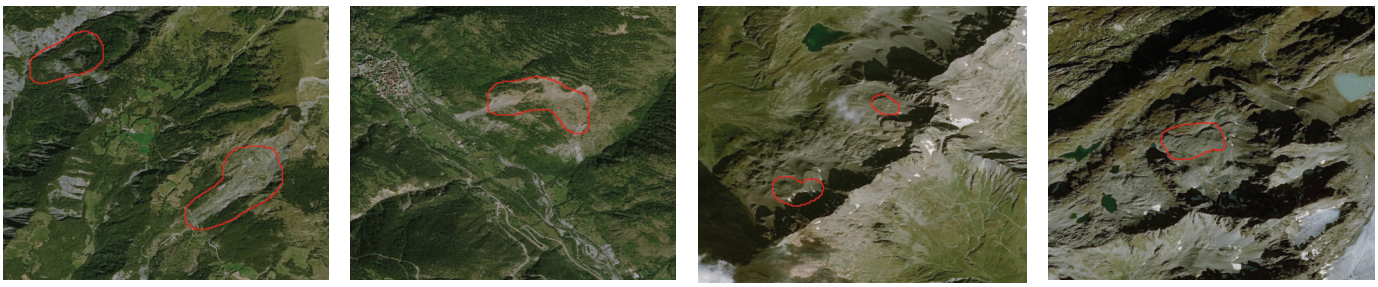
As illustrated in Fig. 6, the entire pipeline used for the detection consists of four steps.

1) ECSPLAIN With a Sliding Window: Due to memory constraints, an entire interferogram cannot be fed directly as input to the network. Therefore, a sliding window strategy is applied over the 54 interferograms. A 100×100 pixels window with an overlap of 1/4 (i.e., a stride of 25 pixels) along each dimension is used to extract the patches to be tested. Each one is provided to the ECSPLAIN framework trained with Grad-CAM to generate a decision and localization map in case of positive match.

2) Stack Average: Once all the interferograms with the same temporal baseline have been processed, the generated localization maps are averaged to ensure consistency in the detection. This strategy is particularly useful for the detection of rock glaciers, which move continuously in time. Conversely, slow-moving landslides can have periods of deactivation that break the continuity of the detection from one interferogram to another. The current step results in three different average localization maps: one with the movements detected with a 6-day temporal baseline, another with a 12-day baseline, and the last with an 18-day baseline.

3) Thresholding: To obtain shapefiles with the polygons of the most susceptible moving areas, a threshold is applied to each average localization map. Based on Table II, the best localization threshold for Grad-CAM is 0.3 for a single interferogram. To mitigate the impact of landslide deactivation periods on the detection and the potential lack of coherence of given regions in certain interferograms (e.g., due to seasonal snow in October), the average localization map is thresholded by 0.1. This threshold is chosen to ensure that the motion is

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(a) Landslides of Barcelonnette (b) Landslide of La Clapière (c) Rock Glaciers near the Col de la Tufe (d) Rock Glacier near the Homme de Tachuy

Fig. 7. Examples of delineation of landslides in (a) and (b), and rock glaciers in (c) and (d) obtained, following the pipeline in Fig. 6. Optical images were acquired in 2018 by the SPOT 6–7 satellites.

TABLE IV

RESULTS OF THE MANUAL VALIDATION STEP OF THE DETECTIONS MADE BY ECSPLAIN. PRECISION IS GIVEN BY ACCOUNTING OR NOT THE LAYOVERS, WHICH CAN BE REMOVED EASILY BY COMPUTING A GEOMETRICAL DISTORTION MAP

Temporal Baseline	6 days			12 days			18 days		
Area ($\times 1.000 m^2$)	[10-50]	[50-100]	100<	[10-50]	[50-100]	100<	[10-50]	[50-100]	100<
Correct	140	80	69	169	86	75	141	77	72
False Positives	67	18	7	52	13	6	76	7	5
Layovers	101	20	13	61	22	11	35	12	5
Total	308	118	89	282	121	92	252	96	82
Precision w. Layovers	0.45	0.68	0.78	0.60	0.71	0.82	0.56	0.80	0.88
Precision w/o Layovers	0.68	0.82	0.91	0.73	0.87	0.93	0.65	0.92	0.94

accurately detected on at least six different interferograms with the same temporal baseline.

4) Filtering: Since this work targets the enrichment of the ISSLIDE dataset, all the detections that intersect already annotated instabilities are not considered. To stay consistent with the annotations already provided in the ISSLIDE dataset, all the detected motions with an area below 10 000 m^2 are also removed in the final map. Finally, an additional check consists of deleting all the instabilities entirely located within distorted areas or near the boundaries of the interferogram.

Following these different steps, three localization maps with the most susceptible moving areas and the shapefiles with the polygons of their location are extracted.

B. Detected Instabilities

Following the pipeline described above, Fig. 7 gives examples of the detections of both slow-moving landslides and rock glaciers. Each detection has been manually validated to evaluate the inference accuracy of the network and identify misleading counterfactual patterns within interferograms. Table IV summarizes the number of correct detections, false positives, and layover-related detections identified by the network depending on their temporal baseline and spatial area. The framework identified a total of 831 unique polygons, among which 479 are correct detections, 163 are false positives, and 189 are related to layover areas. Three main conclusions can be drawn from this performance.

First, the detection pipeline is sensitive to both the temporal baseline and the size of the moving area. As visible in Table IV, the pipeline is more efficient at detecting large moving areas with greater temporal baselines, achieving a precision of 0.88 (last column) compared to 0.45 for small areas with smaller temporal baselines (first column). The size of the detected area critically impacts the reliability of the detection. The pipeline achieves a precision greater than 0.68 for all areas bigger than 50.000 m^2, independently of the baseline. This phenomenon was expected because the coherence map is better exploited on larger areas, and bubble patterns can be caused by small atmospheric delays. The temporal baseline also impacts the precision of the network, with a benefit of more than 0.1 point between the 6-day and 18-day temporal baselines, independently of the size of the detected area. Each temporal baseline targets a different type of phenomenon. Small baselines (6-day) can accurately address the detection of "fast" moving areas, while large baselines (18-day) are less suitable. These areas appear as more noisy patterns on the large temporal baseline interferograms, mostly because of a decrease in coherence. Among the 289 and 290 accurate moving areas detected on the 6-day and the 18-day temporal baselines, respectively, only 138 are common to both baselines, achieving a total of 442 unique correct detections. Therefore, both small and large baselines are necessary to ensure better coverage of the velocity range of slope instabilities.

Second, the major counterfactual patterns identified by the network as moving areas are side effects produced by layover areas. As visible in Fig. 8, the network identifies the bordering regions of layovers as susceptible moving areas, which causes a total of 188 unique misdetections. However, this issue is not critical in our study because a more advanced layover-shadow map could be used, or an overlapping threshold could be set during the filtering step of the pipeline to eliminate

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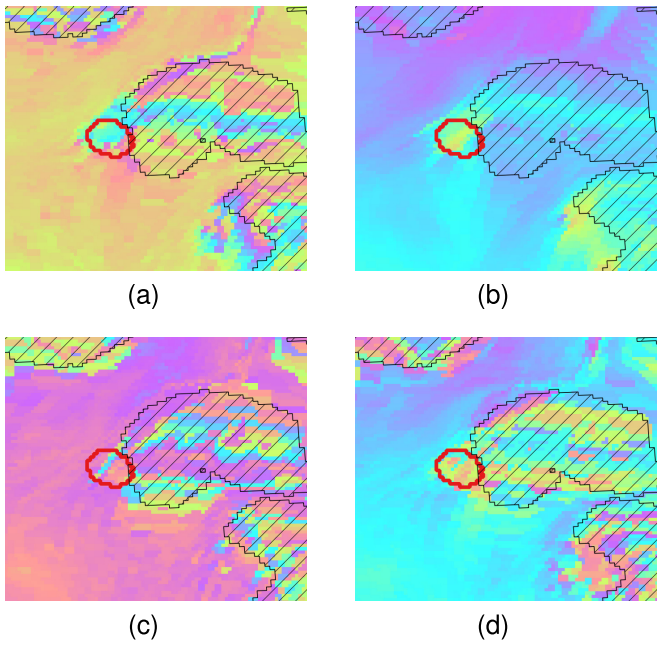


Fig. 8. Misdetection related to layover side effect (red delineated) made by the pipeline in Fig. 6 on four 6-day Sentinel-1 interferograms. Hatching areas refer to layovers. (a) 24/09/2018–30/09/2018. (b) 30/09/2018–06/10/2018. (c) 06/10/2018–12/10/2018. (d) 12/10/2018–18/10/2018.

all these misdetections. It is worth mentioning that, in the current study, we used a layover-shadow map computed as in [80]. This is the default product output by the Sentinel Application Platform (SNAP).[2] We also chose to only discard the detections located entirely within distorted areas to better evaluate the robustness of the network to these layover side effects. Based on the performance in Table IV, removing these layover detections allows the network to achieve a precision greater than 0.7 for most configurations and greater than 0.9 for the largest detected slope instabilities. The other false positives in the table are mainly located in inconsistent areas (e.g., urban regions, highways, bridges, and lakes). These false positives could easily be discarded by including a land cover map or an optical image in the post-processing filtering step.

Finally, among the 479 newly detected moving areas, a large number of them are consistent across the temporal baselines. Specifically, 197 instabilities are common to the 6- and 12-day baselines, 138 are common to the 6- and 18-day baselines, 227 are common to the 12- and 18-day baselines, and 132 are common to all baselines. This consistency strengthens the reliability of the detections and reinforces the relevance of the guidelines proposed in [13]. It also reveals the complementarity of the baselines because 86, 38, and 57 instabilities are only detected in the 6-day, 12-day, and 18-day temporal baselines, respectively.

For the sake of open science research, the resulting localization maps and extracted polygons are made available to future users as additional data in the ISSLIDE dataset on IEEE Dataport (https://ieee-dataport.org/documents/isslide-insar-dataset-slow-sliding-area-detection-machine-learning). All of the correct, false positives, and layover-related detections are provided to encourage taking the difficulties of the network into account during its training. Different deep learning training strategies can be implemented to exploit these polygons as follows.

1) *Updated explainability constraint:* The guidance proposed in the ECSPLAIN framework can be updated in two ways: 1) using a larger loss weight when a counterfactual pattern is input to the network or 2) modify the ResNet50 architecture to output three classes: "Moving Areas"; "Layover side effect"; and "No Move." This would force the network to distinguish between layover-related areas and actual instabilities.
2) *Curriculum learning:* This setting consists of making the task of the network more complex from one training epoch to another. The counterfactual patterns can be used as "difficult" samples that can be added to the dataset later in the training. This will progressively train the network to identify and distinguish slope instability patterns from counterfactuals.
3) *Contrastive learning:* This strategy forces the network to extract different features from a patch of slope instability and a counterfactual patch. By forcing the features to be different, the confusion of the network is reduced, which facilitates the identification of the phenomena of interest.
4) *Weakly supervised learning:* At the time of writing, the polygons are only classified as correct or incorrect detections, but the polygons have not been corrected. Therefore, the provided labels can be used to compute L_{Detect} in (4) without accounting for the annotation map. The annotation map can also be used as an uncertain annotation for computing L_{Loc}.

IX. Conclusion and Perspectives

In this work, a novel Explainability-Constrained claSsifier for Pairing the detection and the Localization of moving Areas on INterferograms (ECSPLAIN) is introduced. This is the first framework that forces a neural network to be explainable and to make decisions based on human-like justifications for detecting moving areas in InSAR data. To this end, the CAM of the classifier is computed at training time and matched to the segmentation of the moving area to make the explanation consistent with the detection. Different CAMs are implemented to ensure the relevance and extensibility of the framework. By applying ECSPLAIN on the ISSLIDE dataset, experiments show that it significantly outperforms standard a posteriori explanation CAMs and shows competitive results against segmentation-only networks. In addition to its highly encouraging performance, the ECSPLAIN framework has several advantages. First, it improves the understandability of InSAR data and network behavior for non-experts by localizing the regions that support the final decision. Second, it reduces computational time, complexity, and memory costs. Finally, it can generate a map of susceptible threats despite misdetections. Our framework is also used for the detection of moving areas on the entire raw interferograms of the ISSLIDE dataset. More than 800 unique susceptible instabilities are

[2]https://step.esa.int/main/toolboxes/snap/

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identified and mapped with ECSPLAIN trained with the Grad-CAM. Among these detections, more than 470 have been verified as actual moving areas. Both the localization maps and the resulting annotation polygons are included with the ISSLIDE dataset as an additional source of annotations for future users.

This ECSPLAIN strategy opens new gates for object detection and localization based on lightweight, rapid, and explainable approaches. Such approaches better fit standard requirements in geoscience for the development of early warning systems and could be extended to the detection of other natural disasters (e.g., volcanoes, earthquakes). To further improve the performance and precision of the classifier network, future works can involve architectural changes, multitemporality, and multimodality. From an architectural point of view, the use of Transformer networks (e.g., SegFormer [81]) has shown strong performance in terms of segmentation. Based on such an assumption, it would be relevant to compare segmentation Transformers with classification Transformers trained under an ECSPLAIN-like framework. This idea would leverage the strong generalizability of attention mechanisms and ensure lighter and explainable decisions. From a multitemporal point of view, the ISSLIDE dataset provides both successive interferograms with the same temporal baseline and multiple interferograms with different temporal baselines but with the same reference date. Therefore, the implementation of a multitemporal approach would strengthen the reliability of the detections as the continuity of the motion in time could be provided to the network. A multibaseline strategy could also be used to make the matches of the detected patterns consistent from one baseline to another. From a multimodal point of view, the digital elevation model or an annual optical image could be used as an additional source of information to validate the detections based on external factors (i.e., the underlying slope and/or geomorphological surface clues). Since all of these approaches are based on feeding a network with multiple inputs, further research can also be conducted in terms of explainability to understand how the inputs combine with each other. Finally, it is worth mentioning that a large amount of InSAR data is made available thanks to online platforms (e.g., FLATSIM service [27]). But, due to the lack of labels, they cannot be exploited in the current study. Further works on weakly-supervised or unsupervised approaches could be further considered to benefit from this amount of available data.

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